

Automatic Video Highlight Detection and Captioning

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Motivation & Introduction

Why video tasks matter?

The exponential growth of digital video content—spanning **social media**, **sports broadcasting**, and **surveillance**—has created a critical need for automated content distillation. Video Highlight Detection (VHD) and Video Captioning (VC) represent two fundamental pillars of video understanding:

- **Highlight Detection:** Aiming to temporally localize segments that are most informative, exciting, or representative of the overall video.
- **Video Captioning:** Tasked with generating natural language descriptions that capture the semantic essence of these segments.

Why is the synergy between Detection and Captioning essential?

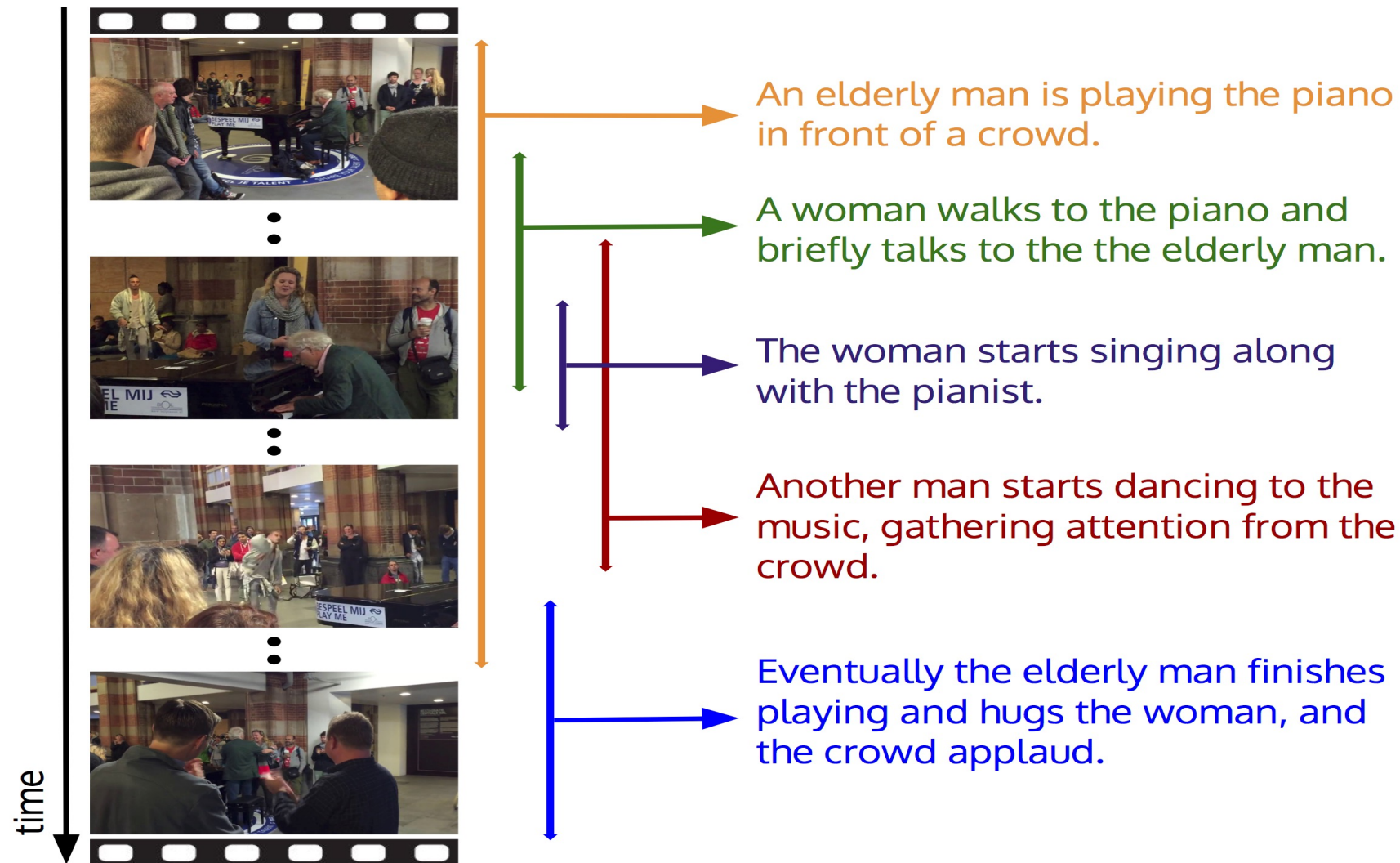
- The "Data Deluge" Challenge
- Enhancing User Experience (UX)
- Real-World Applications



Dataset

ActivityNet Captions is a large-scale benchmark designed for the task of Dense Video Captioning. Unlike standard video captioning, which provides a single summary for an entire clip, It requires models to both temporally localize and describe multiple distinct events within a long-form video.

- **Source:** Built upon the ActivityNet v1.3 hierarchy, covering a wide range of complex human activities.
- **Scale:** Contains 20k videos totaling over 849 hours of footage.
- **Density:** Approximately 100k total descriptions, with an average of 3.65 events per video.
- **Diversity:** Features a large vocabulary and diverse action categories, from sports to daily household chores.



The dataset challenges models to solve two sub-problems simultaneously:

- **Event Localization:** Identifying the start and end timestamps (starti,endi) for all meaningful events.
- **Event Description:** Generating a coherent natural language sentence (Si) for each localized segment.

Key References

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- [2] T. Lin, X. Liu, X. Li, E. Ding, and S. Wen, "BMN: Boundary-matching network for temporal action proposal generation," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, 2019, pp. 3889–3898.
- [3] T. Lin, X. Zhao, H. Su, C. Wang, and M. Yang, "BSN: Boundary sensitive network for temporal action proposal generation," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 3–19.
- [4] H. Liu, C. Li, Q. Wu, and Y. J. Lee, "Visual instruction tuning," *arXiv preprint arXiv:2304.08485*, 2023.
- [5] J. Li, D. Li, C. Xiong, and S. Hoi, "BLIP: Bootstrapping language-image pre-training for unified vision-language understanding and generation," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2022.

Visual Feature Extraction

The temporal segment network models a sequence of snippets **T** as follows:

$$TSN(T_1, T_2, \dots, T_K) = \mathcal{H}(\mathcal{G}(\mathcal{F}(T_1; W), \mathcal{F}(T_2; W), \dots, \mathcal{F}(T_K; W)))$$

Where **F** is convolutional network, **G** is Segmental Consensus Function, **H** is Softmax.

Segmental Consensus Function **G**

- Max Pooling

$$g_i = \max_{k \in \{1, 2, \dots, K\}} f_i^k$$

- Top-K Pooling

$$g_i = \frac{1}{K} \sum_{k=1}^K \alpha_k f_i^k$$

- Average Pooling

$$g_i = \frac{1}{K} \sum_{k=1}^K f_i^k$$

- Linear Weighting

$$g_i = \sum_{k=1}^K \omega_k f_i^k$$

- Attention Weighting

$$e_k = \omega^{att} \mathcal{R}(T_k) \quad \mathcal{A}(T_k) = \frac{\exp(e_k)}{\sum_{i=1}^K \exp(e_i)} \quad g_i = \sum_{k=1}^K \mathcal{A}(T_k) f_i^k$$

- Class score for sliding window

$$F_i^{s,l} = \max_{p \in \{1, 2, \dots, l\}} \{f_i^{s+p}\}$$

Localization

Sampling Mask Weight

$$w_{i,j,n}[t] = \begin{cases} 1 - dec(t_n) & \text{if } t = floor(t_n) \\ dec(t_n) & \text{if } t = floor(t_n) + 1 \\ 0 & \text{if } t = others \end{cases} \quad \text{Extract feature in Non-Integer time}$$

Feature Sampling Dot Product

$$m_{i,j}^f[c, n] = \sum_{t=1}^T S_f[c, t] \cdot w_{i,j}[n, t]. \quad \text{Boundary Matching}$$

Loss of Temporal Evaluation Module

$$L_{TEM} = L_{bl}(P_S, G_S) + L_{bl}(P_E, G_E). \quad \text{Boundary Detection}$$

Weighted Binary Logistic Regression Loss

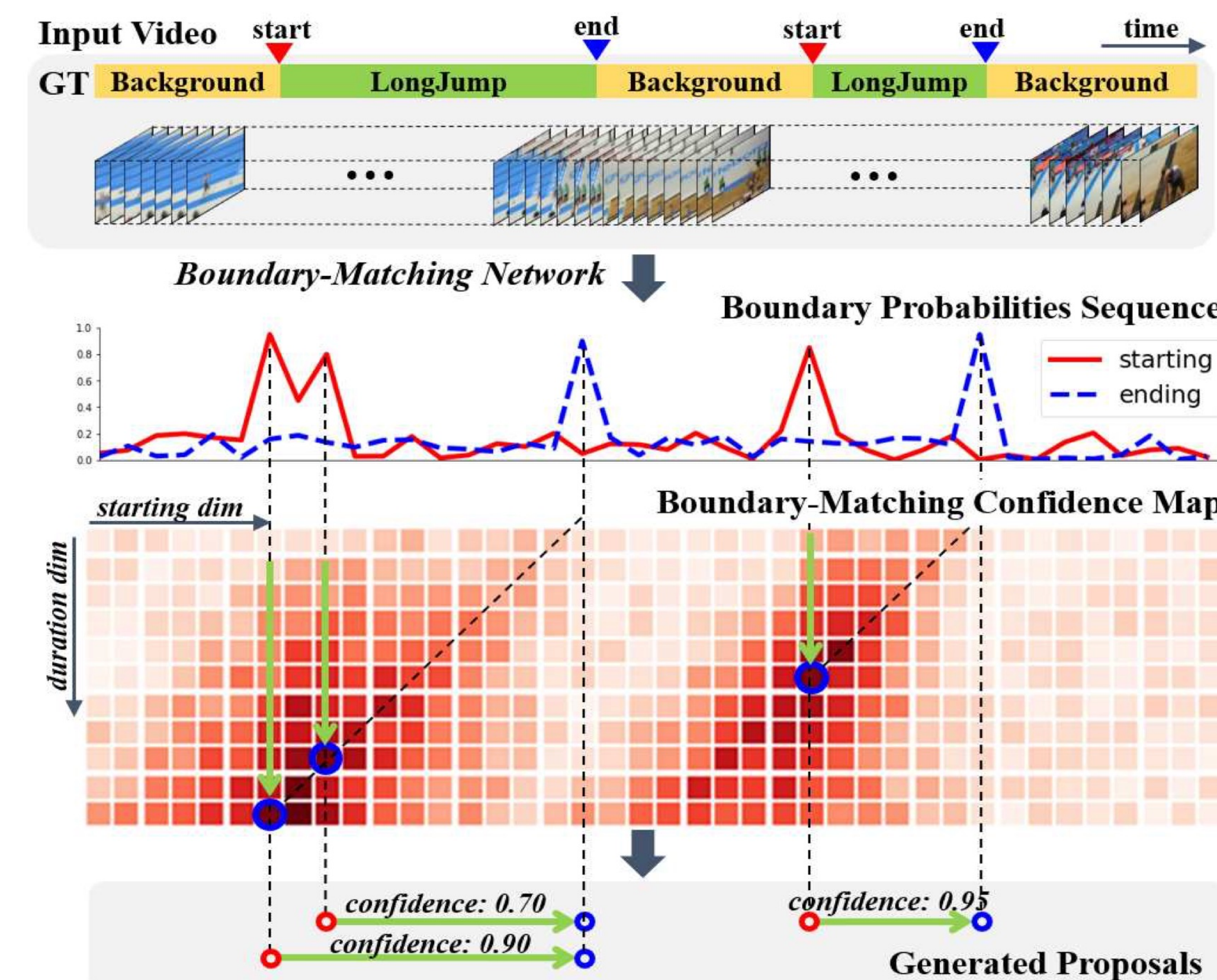
$$L_{bl}(P, G) = \frac{1}{l_w} \sum_{i=1}^{l_w} (\alpha^+ \cdot b_i \cdot \log(p_i) + \alpha^- \cdot (1 - b_i) \cdot \log(1 - p_i)) \quad \text{Balance Negative and Positive Sample}$$

Loss of Proposal Evaluation Module

$$L_{PEM} = L_C(M_{CC}, G_C) + \lambda \cdot L_R(M_{CR}, G_C). \quad \text{Classification and Localization}$$

Training Objective

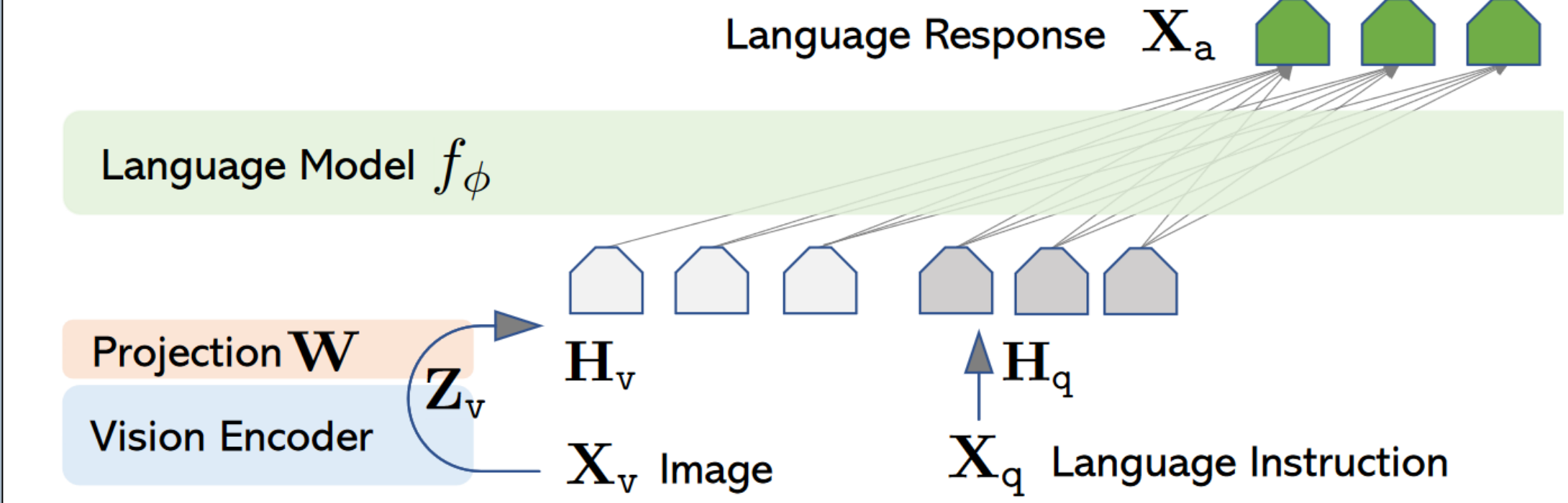
$$L = L_{TEM} + \lambda_1 \cdot L_{PEM} + \lambda_2 \cdot L_2(\Theta).$$



Captioning

We deploy **LLaVA** as our Video Captioning Module. LLaVa is an end-to-end trained large multimodal model that connects a vision encoder and an LLM for general-purpose visual and language understanding. It demonstrates impressive multimodal chat abilities.

LLaVA network architecture.



Word embedding space

$$\mathbf{H}_v = \mathbf{W} \cdot \mathbf{Z}_v, \text{ with } \mathbf{Z}_v = g(\mathbf{X}_v)$$

Which **X** is input image, **g** is ViT/L-14 encoder **Z** is visual feature, **W** is trainable projection matrix.

Instruction at t-th turn

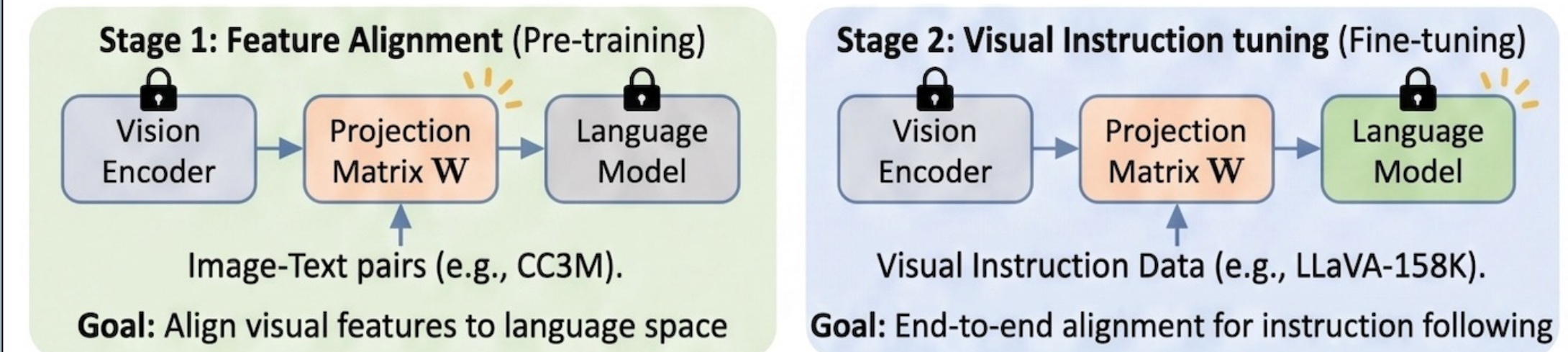
$$\mathbf{X}_{\text{instruct}}^t = \begin{cases} \text{Randomly choose } [\mathbf{X}_q^1, \mathbf{X}_v] \text{ or } [\mathbf{X}_v, \mathbf{X}_q^1], & \text{the first turn } t = 1 \\ \mathbf{X}_q^t, & \text{the remaining turns } t > 1 \end{cases}$$

probability of the target answers **X_answer**

$$p(\mathbf{X}_a | \mathbf{X}_v, \mathbf{X}_{\text{instruct}}) = \prod_{i=1}^L p_{\theta}(\mathbf{x}_i | \mathbf{X}_v, \mathbf{X}_{\text{instruct}, < i}, \mathbf{X}_a, < i).$$

L is length of a sequence

Training Strategy



Evaluation

BMN

Extraction Method	AUC	AR@1	AR@5	AR@10	AR@100
CUHK_Mean_100	67.25	32.89	49.43	56.64	75.29
Slowonly-K700	68.04	33.44	50.53	57.65	75.77

BSN

CUHK_Mean_100	66.26	32.71	48.43	55.28	74.27
Slowonly-K700	67.63	33.04	48.79	56.01	75.74

AUC: Area Under the Average Recall curve, representing the overall quality of generated proposals.
AR@x: Average Recall calculated at x proposals per video, measuring the retrieval coverage within a fixed budget.



- "BLIP-Base": "a group of young people standing on top of a wooden floor"
- "BLIP2-3.7b": "three people dancing in a dance studio"
- "MiniGPT4-7b": "The image shows three people performing a yoga class in a gym. They are all wearing white t-shirts and black shorts, with their arms raised above their heads. One of the girls is holding onto a rope while she lifts her leg up towards the ceiling. Another girl has her hands behind her neck as she reaches for her toes. The third girl has her head down and appears to be doing some poses on the ground. There are no props or any other equipment visible in the background."
- "LLaVA-7b-v1.5": "In the image, there are three people standing in a room, each wearing a white shirt. One of the individuals is a man wearing a white shirt and shorts, while the other two are women wearing white shirts"
- "LLaVA-7b-v1": "The image features a group of people standing in a gym, with one person wearing a shirt that says \"I love UW\" and another person wearing a shirt that says \"I love UW.\" They are standing next to"

Conclusion

- **Unified Framework:** Successfully coupled temporal importance with semantic storytelling.
- **Scalability:** Demonstrated robust performance across diverse domains (Sports, News, Social Media).
- **Impact:** Enhanced content discoverability and accessibility for large-scale video repositories.